Agentic MMIX Pipeline — Low-Level Design & System Documentation

Part 3: Feature Registry & Model Architecture

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1. Raw Data Features

The pipeline ingests 7 CSV files containing the following key variables used in modeling:

1.1 Target Variable

Feature	Source File	Type	Range (Monthly)	Description
total_gmv	SecondFile.csv / firstfile.csv	Continuous (₹)	18 Cr – 51 Cr	Gross Merchandise Value — total revenue. Primary target for all MMIX models.

1.2 Media Channel Spend (9 Channels)

Channel	Source	Monthly Range (₹)	GMV Correlation	Group Assignment
TV	SecondFile.csv	Variable	+0.683 (strong)	traditional
Sponsorship	SecondFile.csv	Variable	+0.726 (strong)	traditional
Online.marketing	SecondFile.csv	Variable	+0.887 (strong)	digital_performance
Affiliates	SecondFile.csv	Variable	+0.866 (strong)	digital_performance
Digital	SecondFile.csv	Variable	+0.390 (moderate)	digital_brand
Content.Marketing	SecondFile.csv	Variable	+0.617 (strong)	digital_brand
SEM	SecondFile.csv	Variable	+0.535 (moderate)	digital_brand
Radio	SecondFile.csv	Variable	-0.946 (strong neg)	other
Other	SecondFile.csv	Variable	-0.959 (strong neg)	other

1.3 Control & Contextual Variables

Feature	Source	Type	Range	Description
NPS	MonthlyNPSscore.csv	Continuous	44–60	Net Promoter Score. Correlation with GMV: -0.96. Used as seasonality proxy (A4), not customer satisfaction measure.
total_Discount	SecondFile.csv	Continuous (₹)	Variable	Total discount amount across all products
total_Mrp	SecondFile.csv	Continuous (₹)	Variable	Total MRP (list price) — denominator for discount ratio
Total.Investment	SecondFile.csv	Continuous (₹)	Variable	Sum of all 9 channel spends
sale_flag	SpecialSale.csv	Binary (0/1)	0–1	Whether any sale event occurred in this period
sale_days	SpecialSale.csv	Integer	0–7	Number of unique sale dates in the period
sale_intensity	SpecialSale.csv	Integer	0–3	Number of unique sale event names in the period

2. Engineered Features

2.1 Log Transforms (12 variables)

All spend-related variables are log-transformed using $\log(x + 1)$ to enable elasticity interpretation and handle zero-spend periods cleanly.

Raw Column	Log Column	Purpose
total_gmv	**log_total_gmv**	Model target in log-log and log-linear specs
TV	log_TV	Channel elasticity
Digital	log_Digital	Channel elasticity
Sponsorship	log_Sponsorship	Channel elasticity
Content.Marketing	log_Content.Marketing	Channel elasticity
Online.marketing	log_Online.marketing	Channel elasticity
Affiliates	log_Affiliates	Channel elasticity
SEM	log_SEM	Channel elasticity
Radio	log_Radio	Channel elasticity
Other	log_Other	Channel elasticity
Total.Investment	log_Total_Investment	Aggregate spend elasticity
total_Discount	log_total_Discount	Discount elasticity

Why log(x+1)? The +1 ensures log(0) doesn't produce -∞ for zero-spend periods. In a log-log model, coefficients are directly interpretable as elasticities: a 1% increase in spend produces a β% change in GMV.

2.2 Channel Group Spend (8 variables)

Individual channels are aggregated into 4 groups based on **EDA correlation analysis** (not business convention alone). Both raw and log-transformed versions are created.

Group	Constituent Channels	Raw Column	Log Column
traditional	TV + Sponsorship	spend_traditional	log_spend_traditional
digital_performance	Online.marketing + Affiliates	spend_digital_performance	log_spend_digital_performance
digital_brand	Digital + Content.Marketing + SEM	spend_digital_brand	log_spend_digital_brand
other	Radio + Other	spend_other	log_spend_other

2.3 Channel Group Validation (from EDA)

The grouping was validated through three levels of correlation analysis:

Level 1 — Intra-Group Correlations (must be high):

Group	Channel Pair	Correlation	Assessment
digital_performance	Online.marketing ↔ Affiliates	**r = 0.989**	Must group — essentially the same signal
digital_brand	Digital ↔ Content.Marketing	r = 0.900	Strong — same strategic category
digital_brand	Digital ↔ SEM	**r = 0.973**	Very strong — SEM moved here from digital_perf
digital_brand	Content.Marketing ↔ SEM	r = 0.958	Strong — confirms grouping
traditional	TV ↔ Sponsorship	r = 0.550	Moderate — distinct enough for separate specs but same strategic category

Level 2 — Inter-Group Correlations (must not be too high):

Group Pair	Group-Level Correlation	Assessment
traditional ↔ digital_performance	0.513	OK — well below 0.7
traditional ↔ digital_brand	0.746	Borderline but acceptable — different strategies
digital_performance ↔ digital_brand	0.386	OK — distinct signals
traditional ↔ other	-0.169	OK — uncorrelated
digital_performance ↔ other	0.335	OK
digital_brand ↔ other	-0.174	OK

Level 3 — Group-GMV Correlations:

Group	GMV Correlation	Interpretation
digital_performance	**+0.884**	Strongest revenue driver
traditional	+0.751	Strong second
digital_brand	+0.507	Moderate — but convergence shows negative marginal impact
other	+0.214	Weak (Radio/Other individually negative)

SEM Reassignment Decision: SEM was originally in `digital_performance`. EDA revealed SEM correlates with Digital (r=0.97) and Content.Marketing (r=0.96) far more strongly than with Online.marketing (r=0.44). SEM was moved to `digital_brand` to reflect actual co-movement patterns.

2.4 Promotional Features

Feature	Calculation	Type	Interpretation
sale_flag	Binary: any sale event in period	0/1	Promotional period indicator
sale_days	Count of unique sale dates per period	Integer (0-7)	Promotional duration
sale_intensity	Count of unique sale event names per period	Integer (0-3)	Promotional depth
discount_intensity	total_Discount / total_Mrp	Ratio (0.38–0.50)	Price reduction depth. Avoids endogeneity from absolute amounts.
discount_per_unit	total_Discount / total_Units	Ratio	Per-unit discount — alternative measure

2.5 Control & Momentum Features

Feature	Calculation	Purpose	Notes
nps_standardized	(NPS − mean) / std	Seasonality proxy (A4)	NOT customer satisfaction. Negative coefficient expected. Removing drops R^2 significantly.
total_gmv_lag1	total_gmv shifted 1 period	Momentum / carryover	Captures whether last period's GMV predicts this period's. Costs 1 observation.
Total.Investment_lag1	Total.Investment shifted 1 period	Spend carryover	Available but not in current specs — reserved for future adstock modeling.

2.6 Seasonality Features (Available, Not in Specs)

Feature	Calculation	Purpose	Status
is_festival_season	1 if month is Oct/Nov/Dec	Festival quarter indicator	Created but NOT in any spec. Was tested as NPS replacement but NPS proved more effective for R^2.
month_sin	$\sin(2\pi \times \text{month} / 12)$	Cyclical month encoding	Created but NOT in any spec. Avoids Dec→Jan discontinuity.
month_cos	$\cos(2\pi \times \text{month} / 12)$	Cyclical month encoding	Created but NOT in any spec. Pair with month_sin for full cycle.

Why not in specs? Testing showed that removing NPS and adding seasonality features produced lower R^2 than keeping NPS. NPS absorbs seasonal variance more effectively because it captures the intensity of seasonal effects (varies month to month), while is_festival_season is binary and month_sin/cos are fixed patterns. NPS is kept with clear documentation that it's a seasonality proxy (A4).

2.7 Feature Matrix Summary

Granularity	Rows	Total Columns	Specs	Features per Spec
Weekly	47	~46	10 (6 base + 4 weekly)	3–8
Monthly	10	~24	6 (base only)	3

Note: The matrix contains ALL features (raw, log, groups, controls, seasonality) — far more columns than any single spec uses. Each spec selects a specific subset.

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3. Model Specifications (10 Curated)

3.1 Design Principles

1. **Each spec tests a distinct hypothesis** — no two specs test the same question
2. **Base specs (A-F):** 3 features maximum, safe for monthly n=10. Tested in iteration 1 ("base" strategy).
3. **Weekly specs (G-J):** 6-8 features, group-based. Only added when n≥30. Tested in iteration 2 ("expanded" strategy, delta only).
4. **No individual channels in weekly specs:** EDA showed inter-channel correlations of r=0.97-0.99, making individual channel coefficients unreliable. Weekly specs use channel groups instead, with additional

control features (discount, lag, NPS) to compensate for fewer channel dimensions.

5. **Channel selection is data-driven:** Spec C uses top-2 channels ranked by |correlation| with log_total_gmv. No hardcoded channel names.
6. **NPS included as seasonality proxy** where needed (specs B, F, G, H, J) per Assumption A4.

3.2 Base Specs (A-F) — Always Available

Spec	Target	Features	Count	Hypothesis
A	log_total_gmv	log_spend_traditional, log_spend_digital_performance, sale_flag	3	Do the two strongest channel groups (traditional + digital_perf) plus sale events explain GMV?
B	log_total_gmv	log_Total_Investment, sale_flag, nps_standardized	3	Does aggregate spend with sale and seasonality control explain GMV? This is the simplest meaningful baseline.
C	log_total_gmv	top2_channels (data-driven), sale_flag	3	Can just the best 2 individual channels (by
D	log_total_gmv	log_spend_digital_performance, sale_flag, total_gmv_lag1	3	Does the strongest channel group with momentum (last period's GMV) predict better than without? Tests carryover effects.
E	log_total_gmv	log_Total_Investment, discount_intensity, sale_flag	3	Does promotional depth (discount ratio) drive revenue beyond what total spend and sale events capture? Tests the discounting lever.
F	log_total_gmv	log_Total_Investment, sale_days, nps_standardized	3	Does the number of sale days matter more than the binary sale flag? Tests whether promotional duration drives GMV beyond just having a sale.

3.3 Weekly Specs (G-J) — Added When n ≥ 30

Spec	Target	Features	Count	Hypothesis
G	log_total_gmv	log_spend_traditional, log_spend_digital_performance, log_spend_digital_brand, sale_flag, discount_intensity, nps_standardized	6	Do the 3 main channel groups with promotional depth and seasonality control explain GMV better than aggregate spend? Tests group-level attribution.

Spec	Target	Features	Count	Hypothesis
H	log_total_gmv	log_spend_traditional, log_spend_digital_performance, log_spend_digital_brand, sale_flag, total_gmv_lag1, nps_standardized	6	Same as G but with momentum instead of discount. Tests whether carryover effects improve group-level models.
I	log_total_gmv	log_spend_traditional, log_spend_digital_performance, log_spend_digital_brand, log_spend_other, sale_flag, discount_intensity, total_gmv_lag1	7	All 4 channel groups with both controls (discount + lag). Tests if including the "other" group (Radio/Other, negative correlation) improves or hurts the model.
J	log_total_gmv	log_spend_traditional, log_spend_digital_performance, log_spend_digital_brand, log_spend_other, sale_flag, discount_intensity, nps_standardized, total_gmv_lag1	8	Maximum information: all 4 groups + all 3 controls. The most complex spec — relies on Ridge/Lasso/ElasticNet to handle multicollinearity and select relevant features.

3.4 Spec-Feature Matrix

Feature	A	B	C	D	E	F	G	H	I	J
log_spend_traditional	✓						✓	✓	✓	✓
log_spend_digital_performance	✓			✓			✓	✓	✓	✓
log_spend_digital_brand							✓	✓	✓	✓
log_spend_other									✓	✓
log_Total_Investment		✓			✓	✓				
top2 channels (dynamic)			✓							
sale_flag	✓	✓	✓	✓	✓		✓	✓	✓	✓
sale_days						✓				
discount_intensity					✓		✓		✓	✓
nps_standardized		✓				✓	✓	✓		✓
total_gmv_lag1				✓				✓	✓	✓

Feature	A	B	C	D	E	F	G	H	I	J
Total features	**3**	**3**	**3**	**3**	**3**	**3**	**6**	**6**	**7**	**8**

3.5 Why No Individual Channels in Weekly Specs

EDA correlation analysis revealed extreme inter-channel correlations that make individual channel coefficients unreliable in multi-channel regressions:

Channel Pair	Correlation	Problem
Online.marketing ↔ Affiliates	**0.989**	Essentially the same signal — OLS cannot separate
SEM ↔ Digital	0.973	Near-identical spending patterns
Content.Marketing ↔ SEM	0.958	Co-movement too strong
Digital ↔ Content.Marketing	0.900	High overlap

With correlations above 0.9, individual channel coefficients become highly unstable — small data perturbations can flip signs and magnitudes. VIF values for specs using multiple individual channels exceeded 1000.

Solution: Weekly specs use channel GROUPS instead. Groups aggregate the correlated channels, producing VIF values under 10 for most specs. Ridge/Lasso provide additional regularization for remaining multicollinearity.

Spec C is the exception — it uses 2 individual channels (data-driven top-2 by |correlation|). This works because only 2 channels are included simultaneously, keeping VIF manageable.

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4. Model Architecture

4.1 Algorithm Types (6 Linear Models)

Model	Library	Type	Key Hyperparameters	Strengths	Weaknesses	Why Included
OLS	statsmodels	Linear	None	P-values, AIC/BIC, full interpretability	No regularization, unstable with multicollinearity	Baseline: standard MMIX approach, statistical significance
Ridge	scikit-learn	Linear	alpha=1.0	L2 regularization,	No feature selection, no p-	Handles group multicollinearity;

Model	Library	Type	Key Hyperparameters	Strengths	Weaknesses	Why Included
				stable coefficients even with correlated features	values	typically the best performer
Lasso	scikit-learn	Linear	alpha=CV-tuned, max_iter=10000	L1 regularization, automatic feature selection (zeros out weak features)	Can zero out important features if correlated with stronger ones	Natural feature selection for high-feature specs (G-J)
ElasticNet	scikit-learn	Linear	alpha=CV, l1_ratio=[0.1-0.9]	Combined L1+L2, balanced regularization	More hyperparameters to tune	Best of both worlds for mixed feature types
Bayesian Ridge	scikit-learn	Linear	Auto-tuned alpha/lambda	Uncertainty estimates, automatic regularization	Slower, no p-values	Confidence intervals on elasticities
Huber	scikit-learn	Linear	epsilon=1.35	Robust to outliers — downweights high-residual points	No p-values, no feature selection	Sale-period GMV spikes are leverage points; Huber reduces their influence

4.2 Why No Tree-Based Models

XGBoost and Random Forest were evaluated but **excluded from the final pipeline** because:

1. **No interpretable elasticities:** Tree models produce feature importance (relative ranking) but not coefficients (directional magnitude). MMIX requires knowing "a 10% increase in Sponsorship spend produces X% GMV lift" — trees cannot provide this.
2. **No coefficient signs:** Trees cannot tell you whether a channel has positive or negative marginal impact. The convergence-based ordinality check requires signed coefficients.
3. **Scenario simulation incompatibility:** The scenario simulator multiplies baseline features by channel change factors and re-predicts. This works naturally with linear models but produces unreliable results with trees (extrapolation outside training data range).

4. **LOO CV overhead:** With n=47 and tree models, LOO CV is slow (47 × full model fit). Linear models are fast.

The builder functions (`build_xgboost`, `build_random_forest`) remain in the codebase for potential future use with larger datasets where non-linear effects become detectable.

4.3 Transform Variants (2)

Variant	Target	Features	Coefficient Interpretation	When Better
log_log	log(GMV+1)	log(spend+1)	**Elasticity:** 1% increase in spend → β% increase in GMV	Standard MMIX interpretation. Handles diminishing returns naturally.
log_linear	log(GMV+1)	raw spend	**Semi-elasticity:** 1 unit (₹) increase in spend → β% increase in GMV	When absolute spend amounts are more interpretable than percentages.

The `LOG_TO_RAW_MAP` in config.py maps log feature names back to raw column names, enabling the modeling pipeline to resolve features correctly for each transform variant.

4.4 Total Model Candidates

Iteration	Strategy	Granularity	Specs	Types	Transforms	Candidates
1	base	weekly	6 (A-F)	6	2	**72**
2	expanded (delta)	weekly	4 (G-J only)	6	2	**48**
3	monthly (fallback)	monthly	6 (A-F)	6	2	**72**
Max total						**192**

Not all candidates succeed — some are skipped due to missing columns, insufficient rows, or build failures. Typical success rates: ~83% for weekly (60/72), ~50% for monthly (36/72 due to n=10 limiting complexity).

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5. Evaluation Metrics

5.1 Composite Score Formula

```
Composite = 0.35 × Fit + 0.25 × Stability + 0.25 × Ordinality + 0.15 × VIF
```

Component	Weight	Metric	Calculation	Why This Weight
Fit	35%	Adjusted R^2	max(0, min(1, adj_r_squared))	Primary: does the model explain GMV? Adjusted penalizes overfitting from extra predictors.
Stability	25%	1 −	Train R^2 − CV R^2	/ Train R^2
Ordinality	25%	Convergence-based (see below)	1.0 if all convergence-confirmed features match; penalized per violation	Business logic: coefficients should align with cross-model consensus.
VIF	15%	Scaled by max VIF	≤5: 1.0, ≤10: 0.7, ≤50: 0.3, >50: 0.1	Multicollinearity reliability. Lower weight because Ridge/Lasso inherently handle it.

5.2 Why These Weights

Metric	Why Chosen	Business Relevance
Adjusted R^2	Penalizes overfitting from extra predictors	With n=47, adding features always improves R^2 but may not generalize
LOO CV	Leave-one-out is the most thorough CV for small samples	Each data point gets a turn as test set — no lucky/unlucky splits
Convergence ordinality	Uses cross-model consensus instead of hardcoded rules	Digital_brand is genuinely negative; hardcoded "must be positive" would penalize correct models
VIF	Multicollinearity detection	Lower weight because regularized models handle it; still useful for flagging extreme cases

5.3 Convergence-Based Ordinality (Two-Pass System)

Traditional MMIX enforces "all spend coefficients must be ≥ 0." This fails for:

- Channels with negative marginal impact when controlling for others (digital_brand)
- Channels with negative raw correlation (Radio, Other)
- NPS (seasonality proxy with expected negative coefficient)

The two-pass system uses cross-model consensus as the truth:

Pass 1 — Build and Rank All Models: Build all (spec × type × transform) candidates. Initial ordinality check only enforces positive on unambiguous features: `sale_flag`, `sale_days`, `Total.Investment`. All channel/group features have no constraint. Rank by composite score.

Pass 2 — Convergence Re-Scoring:

1. Compute convergence across ALL successful models:
   - For each feature: collect coefficients from all models containing it
   - If ≥ 80% of models have positive coefficient → "POSITIVE (confirmed)"
   - If ≤ 20% positive → "NEGATIVE (confirmed)"
   - Otherwise → "MIXED (inconclusive)"
2. Re-score ordinality for EACH model:
   - For each confirmed feature in the model: does the coefficient match the confirmed direction?
   - Match → PASS for that feature
   - Mismatch → VIOLATION
   - Ordinality score = proportion of checks that pass
3. Re-compute composite score with updated ordinality. Re-rank all models.

Example:

Feature	Convergence Direction	Model A Coef	Model A Check	Model B Coef	Model B Check
sale_flag	POSITIVE (confirmed)	+0.22	✅ PASS	+0.18	✅ PASS
log_spend_traditional	POSITIVE (confirmed)	+0.17	✅ PASS	-0.05	❌ FAIL
log_spend_digital_brand	NEGATIVE (confirmed)	-0.15	✅ PASS	+0.03	❌ FAIL
nps_standardized	NEGATIVE (confirmed)	-0.04	✅ PASS	+0.01	❌ FAIL

Model A: 4/4 pass → ordinality score = 1.0
Model B: 1/4 pass → ordinality score = 0.25

This ensures Model A (which correctly captures digital_brand as negative) ranks higher than Model B (which has the "intuitive" but incorrect positive digital_brand coefficient).

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6. Cross-Validation Strategy

6.1 Leave-One-Out (LOO) CV

Why LOO? With n=47 (weekly) or n=10 (monthly), k-fold CV has high variance — which 5 or 10 observations end up in the test fold can dramatically change results. LOO eliminates this randomness: every observation gets exactly one turn as the test point.

Process: For each of n observations:

1. Remove observation i from training data
2. Fit model on remaining n-1 points
3. Predict observation i
4. Record prediction

After all n rounds:

- CV R^2 = R^2 of actual vs all held-out predictions
- CV RMSE = RMSE of held-out predictions
- CV MAPE = Mean Absolute Percentage Error of held-out predictions

Cost: n model fits per candidate. For 72 candidates × 47 LOO folds = 3,384 model fits per iteration. Linear models make this feasible (~2-3 minutes total).

6.2 Stability Score

$$\text{Stability} = \max(0, 1 - |\text{Train_}R^2 - \text{CV_}R^2| / \text{Train_}R^2)$$

A model with Train R^2 = 0.58 and CV R^2 = 0.36 has stability = 1 - |0.58-0.36|/0.58 = 0.62. This is moderate — the model explains less on held-out data than training data, indicating some overfitting.

Perfect stability (1.0) means CV R^2 equals Train R^2. Zero stability means CV R^2 is zero or negative.

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7. Scenario Simulation Architecture

7.1 Simulator Design

The scenario simulator translates raw business decisions ("increase Sponsorship by 20%") into model-level feature changes, predicts the outcome, and converts back to business metrics.

Translation chain:

```
Raw change: {"Sponsorship": 1.2}
   ↓ (if log-log transform)
Feature change: log(Sponsorship × 1.2 + 1) - log(Sponsorship + 1)
   ↓ (if model uses groups)
Group update: spend_traditional = TV + (Sponsorship × 1.2)
   ↓
log_spend_traditional = log(new_spend_traditional + 1)
   ↓
Model prediction → log(GMV_predicted)
   ↓
GMV_predicted = exp(log_pred) - 1 (if log target)
   ↓
Change % = (GMV_predicted - GMV_baseline) / GMV_baseline × 100
```

7.2 Standard Scenarios (11+)

Category	Scenario	Changes Applied
Individual channels	+20% Online.marketing	Online.marketing × 1.2
	+20% Sponsorship	Sponsorship × 1.2
	+20% TV	TV × 1.2
	+20% SEM	SEM × 1.2
	+20% Affiliates	Affiliates × 1.2
	+20% Digital	Digital × 1.2
Budget reallocation	Shift 10% TV → Online.mkt	TV × 0.9, Online.mkt × 1.1
Scale	All channels +10%	All × 1.1
	All channels +20%	All × 1.2
	All channels -20%	All × 0.8
Promotional	Sale event activation	sale_flag = 1, sale_days = 3
	Sale + 10% all	sale_flag = 1 + all × 1.1

7.3 When Scenarios Run

Scenarios are **only run after the agentic loop approves the model** (Phase 8, post-loop). During the iterative loop (Phase 6), `skip_scenarios=True` avoids wasting compute on models that may be rejected.

This means scenarios always reflect the best available model, and the scenario plot (`scenarios.png`) shows results from the final approved model, not an intermediate iteration.